An Overview of Apache Cassandra

This article explores Apache Cassandra, an open source distributed database management system that is capable of handling large amounts of data across many servers with high availability and no single point of failure. Initially developed by Facebook, it is now used by many organisations like Twitter, Reddit, etc.

According to Wikipedia, NoSQL (Not only SQL) databases provide a mechanism for storage and retrieval of data that is modelled on means other than the tabular relations used in relational databases.

There are four main types of NoSQL databases as shown in Figure 1.

Apache Cassandra is an open source NoSQL database and management system. It was designed to handle large amounts of data across many servers, data centres and clouds. It can store hundreds of terabytes of data, and differs a lot from relational database management systems (RDBMS). Apache Cassandra was developed by the Apache Software Foundation in 2008, is written in the Java language, and supports a cross-platform operating system. Initially developed by Facebook to power its inbox search feature, it is being used by many other organisations, such as Twitter, Reddit, Rackspace, Ciscom, Cloudkick, among others.

Cassandra's architecture is different. It consists of various nodes all over the network. There is no master node concept in Cassandra; every node can serve any request. Each node contains different data, which makes it seem as if it is distributed across all the data centres. It provides automatic data distribution across all nodes, and is a 'ring' or database cluster. The concept of replication can be configured across one or many data centres, and also at multiple cloud availability zones for achieving high availability.

Cassandra provides various features such as scalability, durability, storing duplicate copies of data across all the nodes available in a network, and replication. This means that if any node goes down, other copies of the node's data are available on other machines as failover, which can be used. Thus, Cassandra provides anytime access to the data.

There are various key features in Cassandra that are better than what other databases offer. Cassandra consists of a data model in which rows are partitioned. It offers a schema-free

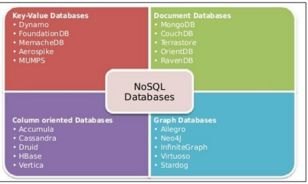

Figure 1: Types of databases